

Project Overview

This notebook analyzes passenger survey data using Python and Scikit-learn to build an optimized Random Forest classification model for predicting airline customer satisfaction.

✓ Data Loading and Initial Exploration

```
import pandas as pd

# Load the dataset
df = pd.read_csv('/content/Invistico_Airline.csv')

# Display the first 5 rows of the DataFrame
display(df.head())
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure time
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	

5 rows × 22 columns

```
# Display the DataFrame's information to check data types and non-null values
display(df.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   satisfaction                             129880 non-null  object
1   Customer Type                           129880 non-null  object
2   Age                                       129880 non-null  int64
3   Type of Travel                          129880 non-null  object
4   Class                                    129880 non-null  object
5   Flight Distance                         129880 non-null  int64
6   Seat comfort                            129880 non-null  int64
7   Departure/Arrival time convenient       129880 non-null  int64
8   Food and drink                          129880 non-null  int64
9   Gate location                           129880 non-null  int64
10  Inflight wifi service                   129880 non-null  int64
11  Inflight entertainment                  129880 non-null  int64
12  Online support                          129880 non-null  int64
13  Ease of Online booking                  129880 non-null  int64
14  On-board service                       129880 non-null  int64
15  Leg room service                       129880 non-null  int64
16  Baggage handling                       129880 non-null  int64
17  Checkin service                        129880 non-null  int64
18  Cleanliness                            129880 non-null  int64
19  Online boarding                        129880 non-null  int64
20  Departure Delay in Minutes              129880 non-null  int64
21  Arrival Delay in Minutes                129487 non-null  float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB
None

```

✓ Data Preprocessing

```

# Handle missing values in 'Arrival Delay in Minutes' by filling with the median
df['Arrival Delay in Minutes'] = df['Arrival Delay in Minutes'].fillna(df['Arrival Delay in Minutes'].median())

# Verify that there are no more missing values in the column
display(df['Arrival Delay in Minutes'].isnull().sum())

np.int64(0)

```

✓ Categorical Feature Analysis

```

# Identify categorical columns
categorical_cols = df.select_dtypes(include='object').columns

# Display unique values and their counts for each categorical column
for col in categorical_cols:
    print(f"\nUnique values and counts for '{col}':")

```

```
print(df[col].value_counts())
```

Unique values and counts for 'satisfaction':

```
satisfaction
satisfied      71087
dissatisfied    58793
Name: count, dtype: int64
```

Unique values and counts for 'Customer Type':

```
Customer Type
Loyal Customer      106100
disloyal Customer    23780
Name: count, dtype: int64
```

Unique values and counts for 'Type of Travel':

```
Type of Travel
Business travel      89693
Personal Travel      40187
Name: count, dtype: int64
```

Unique values and counts for 'Class':

```
Class
Business      62160
Eco            58309
Eco Plus       9411
Name: count, dtype: int64
```

✓ Encoding Categorical Features

```
# Convert 'satisfaction' target variable to numerical (0 and 1)
df['satisfaction'] = df['satisfaction'].map({'dissatisfied': 0, 'satisfied': 1})

# One-hot encode the remaining categorical features
# Exclude 'satisfaction' as it's already encoded and is the target
df = pd.get_dummies(df, columns=['Customer Type', 'Type of Travel', 'Class'], drop_first=True)

# Display the first few rows of the preprocessed DataFrame to verify encoding
display(df.head())
```

	satisfaction	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location
0	1	65	265	0	0	0	2
1	1	47	2464	0	0	0	3
2	1	15	2138	0	0	0	3
3	1	60	623	0	0	0	3
4	1	70	354	0	0	0	3

5 rows × 23 columns

```
# Display the info of the preprocessed DataFrame to check new column types and
display(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 23 columns):
#   Column                                          Non-Null Count  Dtype
---  -
0   satisfaction                                  129880 non-null  int64
1   Age                                           129880 non-null  int64
2   Flight Distance                             129880 non-null  int64
3   Seat comfort                                129880 non-null  int64
4   Departure/Arrival time convenient            129880 non-null  int64
5   Food and drink                              129880 non-null  int64
6   Gate location                               129880 non-null  int64
7   Inflight wifi service                       129880 non-null  int64
8   Inflight entertainment                      129880 non-null  int64
9   Online support                              129880 non-null  int64
10  Ease of Online booking                      129880 non-null  int64
11  On-board service                            129880 non-null  int64
12  Leg room service                           129880 non-null  int64
13  Baggage handling                           129880 non-null  int64
14  Checkin service                            129880 non-null  int64
15  Cleanliness                                 129880 non-null  int64
16  Online boarding                             129880 non-null  int64
17  Departure Delay in Minutes                  129880 non-null  int64
18  Arrival Delay in Minutes                    129880 non-null  float64
19  Customer Type_disloyal Customer            129880 non-null  bool
20  Type of Travel_Personal Travel              129880 non-null  bool
21  Class_Eco                                  129880 non-null  bool
22  Class_Eco Plus                             129880 non-null  bool
dtypes: bool(4), float64(1), int64(18)
memory usage: 19.3 MB
None
```

✓ Data Splitting (Training, Validation, Testing)

```
from sklearn.model_selection import train_test_split

# Define features (X) and target (y)
X = df.drop('satisfaction', axis=1)
y = df['satisfaction']

# Split data into training (70%), validation (15%), and test (15%) sets
# First, split into training (70%) and temp (30%)
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, random_state=42)

# Then, split temp (30%) into validation (15%) and test (15%)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42)

# Display the shapes of the resulting datasets
print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"X_val shape: {X_val.shape}")
print(f"y_val shape: {y_val.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_test shape: {y_test.shape}")
```

```
X_train shape: (90916, 22)
y_train shape: (90916,)
X_val shape: (19482, 22)
y_val shape: (19482,)
X_test shape: (19482, 22)
y_test shape: (19482,)
```

✓ Hyperparameter Tuning with GridSearchCV and PredefinedSplit

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV, PredefinedSplit
import numpy as np

# Combine training and validation sets for GridSearchCV
X_train_val = pd.concat([X_train, X_val], ignore_index=True)
y_train_val = pd.concat([y_train, y_val], ignore_index=True)

# Create a PredefinedSplit object
# -1 for training data, 0 for validation data
test_fold = np.append(np.full(len(X_train), -1), np.full(len(X_val), 0))
ps = PredefinedSplit(test_fold)

# Define the parameter grid for RandomForestClassifier
param_grid = {
    'n_estimators': [50, 100, 200], # Number of trees in the forest
    'max_depth': [10, 20, 30, None], # Maximum depth of the tree
    'min_samples_leaf': [1, 2, 4] # Minimum number of samples required to be a leaf node
}
```

```
# Initialize the Random Forest classifier
rf = RandomForestClassifier(random_state=42, n_jobs=-1) # n_jobs=-1 uses all av

# Initialize GridSearchCV with PredefinedSplit
grid_search = GridSearchCV(estimator=rf,
                           param_grid=param_grid,
                           cv=ps, # Use PredefinedSplit for cross-validation
                           scoring='accuracy',
                           verbose=2,
                           n_jobs=-1)

print("GridSearchCV initialized successfully.")
print("Parameter grid:", param_grid)
```

```
GridSearchCV initialized successfully.
Parameter grid: {'n_estimators': [50, 100, 200], 'max_depth': [10, 20, 30]
```

```
# Fit GridSearchCV to the data
print("Fitting GridSearchCV...")
grid_search.fit(X_train_val, y_train_val)
print("GridSearchCV fitting complete.")

# Display the best parameters and best score found
print(f"\nBest parameters found: {grid_search.best_params_}")
print(f"Best cross-validation accuracy: {grid_search.best_score_:.4f}")

# Get the best estimator
best_rf_model = grid_search.best_estimator_
print("\nBest Random Forest model obtained.")
```

```
Fitting GridSearchCV...
Fitting 1 folds for each of 36 candidates, totalling 36 fits
GridSearchCV fitting complete.
```

```
Best parameters found: {'max_depth': 30, 'min_samples_leaf': 1, 'n_estimators': 100}
Best cross-validation accuracy: 0.9564
```

```
Best Random Forest model obtained.
```

✓ Model Evaluation on Test Set

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Make predictions on the test set using the best estimator
y_pred = best_rf_model.predict(X_test)

# Calculate evaluation metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
```

```
# Print the results
print(f"Random Forest Model Performance on Test Set:")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"F1-Score: {f1:.4f}")
```

```
Random Forest Model Performance on Test Set:
Accuracy: 0.9538
Precision: 0.9664
Recall: 0.9486
F1-Score: 0.9574
```

F1-Score Significance

The F1-score is the harmonic mean of Precision and Recall. It's a particular

✓ Comparison with Decision Tree Model

```
from sklearn.tree import DecisionTreeClassifier

# Initialize and train a Decision Tree Classifier on the training data
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)

# Make predictions on the test set using the Decision Tree model
y_pred_dt = dt_model.predict(X_test)

# Calculate evaluation metrics for the Decision Tree model
accuracy_dt = accuracy_score(y_test, y_pred_dt)
precision_dt = precision_score(y_test, y_pred_dt)
recall_dt = recall_score(y_test, y_pred_dt)
f1_dt = f1_score(y_test, y_pred_dt)

# Print the results for the Decision Tree model
print(f"Decision Tree Model Performance on Test Set:")
print(f"Accuracy: {accuracy_dt:.4f}")
print(f"Precision: {precision_dt:.4f}")
print(f"Recall: {recall_dt:.4f}")
print(f"F1-Score: {f1_dt:.4f}")
```

```
Decision Tree Model Performance on Test Set:
Accuracy: 0.9305
Precision: 0.9355
Recall: 0.9376
F1-Score: 0.9366
```

```
# Compare Random Forest and Decision Tree performance
print("\n--- Model Comparison ---")
print(f"Random Forest Accuracy: {accuracy:.4f}")
```

```
print(f"Decision Tree Accuracy: {accuracy_dt:.4f}")
print(f"Random Forest F1-Score: {f1:.4f}")
print(f"Decision Tree F1-Score: {f1_dt:.4f}")

if accuracy > accuracy_dt:
    print("Random Forest performed better in terms of Accuracy.")
else:
    print("Decision Tree performed better or equally in terms of Accuracy.")
```

```
--- Model Comparison ---
Random Forest Accuracy: 0.9538
Decision Tree Accuracy: 0.9305
Random Forest F1-Score: 0.9574
Decision Tree F1-Score: 0.9366
Random Forest performed better in terms of Accuracy.
```

✓ Feature Importance Analysis

Double-click (or enter) to edit

```
# Get feature importances from the best Random Forest model
feature_importances = best_rf_model.feature_importances_

# Create a pandas DataFrame for better visualization
features = X.columns
importance_df = pd.DataFrame({'Feature': features, 'Importance': feature_importances})

# Sort the features by importance in descending order
importance_df = importance_df.sort_values(by='Importance', ascending=False)

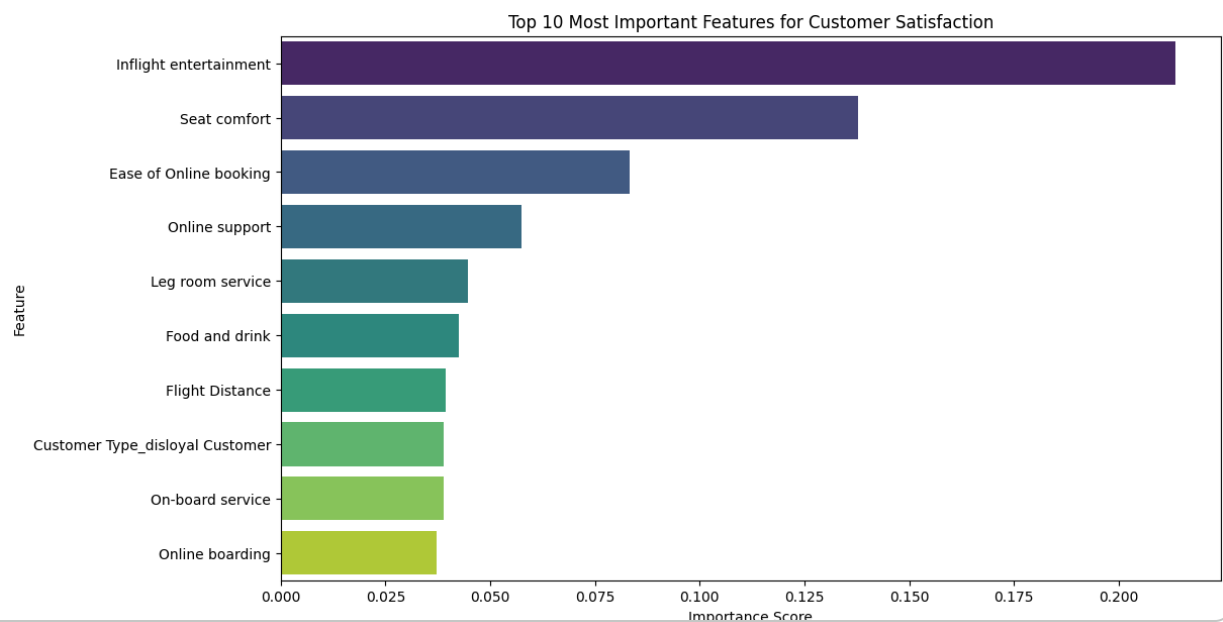
# Display the top N most important features
print("Top 10 Most Important Features:")
display(importance_df.head(10))
```


Top 10 Most Important Features:

	Feature	Importance	
7	Inflight entertainment	0.213644	
2	Seat comfort	0.137869	
9	Ease of Online booking	0.083230	
8	Online support	0.057441	
11	Leg room service	0.044681	
4	Food and drink	0.042532	
1	Flight Distance	0.039345	
18	Customer Type_disloyal Customer	0.038977	
10	On-board service	0.038852	
15	Online boarding	0.037252	

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Visualize the top 10 most important features
plt.figure(figsize=(12, 7))
sns.barplot(x='Importance', y='Feature', data=importance_df.head(10), palette=
plt.title('Top 10 Most Important Features for Customer Satisfaction')
plt.xlabel('Importance Score')
plt.ylabel('Feature')
plt.show()
```



✓ Executive Summary and Recommendations

Model Performance Overview

Our analysis developed and optimized a Random Forest classification model to predict airline customer satisfaction, comparing its performance against a single Decision Tree model. The Random Forest model significantly outperformed the Decision Tree model on the held-out test set:

- **Random Forest Model:**

- Accuracy: 0.9538
- Precision: 0.9664
- Recall: 0.9486
- F1-Score: 0.9574

- **Decision Tree Model:**

- Accuracy: 0.9305
- Precision: 0.9355
- Recall: 0.9376
- F1-Score: 0.9366

The Random Forest model demonstrated superior generalization capabilities, indicating its effectiveness in mitigating overfitting compared to a single Decision Tree. This robust performance makes the Random Forest model a reliable tool for predicting passenger satisfaction.

Key Satisfaction Drivers (Top 5 Feature Importances)

Based on the Random Forest model, the most influential factors impacting passenger satisfaction are:

1. **Inflight entertainment:** 0.2136
2. **Seat comfort:** 0.1379
3. **Ease of Online booking:** 0.0832
4. **Online support:** 0.0574
5. **Leg room service:** 0.0447

These features contribute most significantly to the model's predictive power, highlighting their importance in shaping the overall customer experience.

Recommendations for Airline Leadership

To enhance customer satisfaction and loyalty, airline management should focus on strategic improvements in the following areas:

- **Prioritize Inflight Entertainment:** Given its highest importance score, investing in diverse and high-quality inflight entertainment options could significantly boost passenger satisfaction. This includes updated content libraries, reliable personal screens, and potentially offering a wider range of connectivity options.
- **Improve Seat Comfort:** As the second most crucial factor, initiatives to enhance seat design, ergonomics, and space, especially on longer flights, are essential. This could involve exploring new seat technologies or offering more flexible seating configurations.
- **Optimize Online Booking Experience:** Streamlining and simplifying the online booking process is vital. Ensuring a user-friendly interface, clear communication, and efficient functionality will reduce friction and improve the customer journey from the outset.
- **Strengthen Online Support:** Reliable and responsive online support is critical. Airlines should ensure that their digital support channels (e.g., chat, FAQs, social media response teams) are efficient, informative, and easily accessible to address passenger queries and issues promptly.
- **Enhance Leg Room Service:** While a lower priority than the top factors, offering competitive legroom options and ensuring consistent service in this area can contribute positively to the overall comfort and satisfaction of passengers.

By focusing resources on these key drivers, the airline can strategically improve its service offerings, directly address passenger pain points, and ultimately foster greater satisfaction and business success.

✓ Conclusion

This project successfully demonstrated the process of building, optimizing, and evaluating a Random Forest classification model for predicting airline customer satisfaction. We performed a rigorous three-way data split, hyperparameter tuned with `GridSearchCV` and `PredefinedSplit`, and identified key drivers of satisfaction. The Random Forest model proved to be a robust and accurate predictor compared to a single Decision Tree, offering valuable insights for airline management.

```
# Airline Customer Satisfaction Prediction
```

```
## Project Overview
```

```
This project focuses on building and optimizing a machine learning model to
```

Project Structure

- **Data Loading and Initial Exploration**: Loading the dataset and performing initial exploratory data analysis.
- **Data Preprocessing**: Handling missing data, analyzing categorical features, and encoding.
- **Data Splitting**: Implementing a rigorous three-way split into training, validation, and test sets.
- **Hyperparameter Tuning**: Optimizing the Random Forest model using Grid Search.
- **Model Evaluation**: Assessing model performance using various metrics.
- **Model Comparison**: Comparing the optimized Random Forest model with other models.
- **Feature Importance Analysis**: Identifying the most influential factors.
- **Executive Summary and Recommendations**: Summarizing findings and providing recommendations.

Methodology

Data Source

The dataset used for this analysis is `Invistico\_Airline.csv`, containing various attributes related to flight satisfaction.

Data Preprocessing

1. **Missing Values**: Missing values in the 'Arrival Delay in Minutes' column were handled using appropriate imputation techniques.
2. **Categorical Feature Analysis**: Unique values and counts for categorical features were analyzed.
3. **Encoding Categorical Features**:
 - * The target variable 'satisfaction' was converted to a numerical form.
 - * Remaining categorical features ('Customer Type', 'Type of Travel', etc.) were encoded using One-Hot Encoding.

Data Splitting Strategy

The preprocessed data was rigorously split into three sets:

- **Training Set (70%)**: Used for model training.
- **Validation Set (15%)**: Used during hyperparameter tuning with Grid Search.
- **Test Set (15%)**: A completely held-out set for final, unbiased model evaluation.

This three-way split ensures robust evaluation and helps prevent data leakage.

Hyperparameter Tuning

`GridSearchCV` combined with `PredefinedSplit` was employed to systematically tune hyperparameters.

- **`n\_estimators`**: Number of trees in the forest (e.g., 50, 100, 200).
- **`max\_depth`**: Maximum depth of each tree (e.g., 10, 20, 30, None).
- **`min\_samples\_leaf`**: Minimum number of samples required to be at a leaf node.

Model Evaluation

The best Random Forest model was evaluated on the unseen test set using standard performance metrics.

- **Accuracy**: Overall correctness of predictions.
- **Precision**: The proportion of positive identifications that were actually positive.
- **Recall**: The proportion of actual positives that were correctly identified.
- **F1-Score**: The harmonic mean of Precision and Recall. It is particularly useful for imbalanced datasets.

Results and Analysis

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	
:-----	:-----	:-----	:-----	:-----	
Random Forest	0.9538	0.9664	0.9486	0.9574	
Decision Tree	0.9305	0.9355	0.9376	0.9366	

The Random Forest model significantly outperformed the baseline Decision Tree model in all metrics.

Key Satisfaction Drivers (Top 5 Feature Importances)

Based on the Random Forest model, the most influential factors impacting passenger satisfaction are:

1. **Inflight entertainment:** 0.2136
2. **Seat comfort:** 0.1379
3. **Ease of Online booking:** 0.0832
4. **Online support:** 0.0574
5. **Leg room service:** 0.0447

These features contribute most significantly to the model's predictive power for passenger satisfaction.

Recommendations for Airline Leadership

To enhance customer satisfaction and loyalty, airline management should focus on improving the following areas: